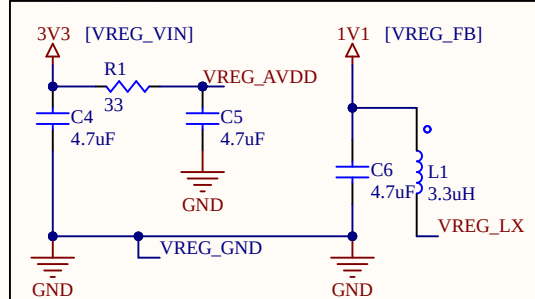
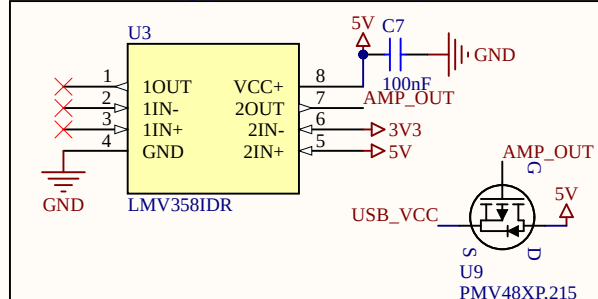


## On-chip voltage regulation circuit [3v3 =&gt; 1v1]

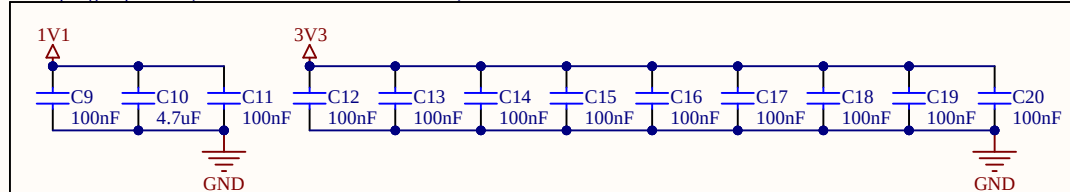


See pg 8 of hardware design manual for PCB layout requirements

## Power selection circuitry (choose VIN or USB\_VCC)

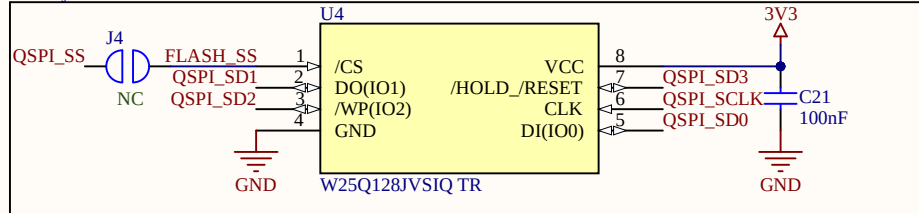


## Decoupling Capacitors (for 1V1 DVDD and 3V3 IOVDD)

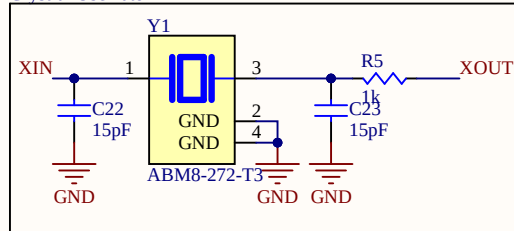


Decoupling caps should be as close to the VDD pins as possible. Because of certain design tradeoffs, some pins may share a decoupling capacitor.

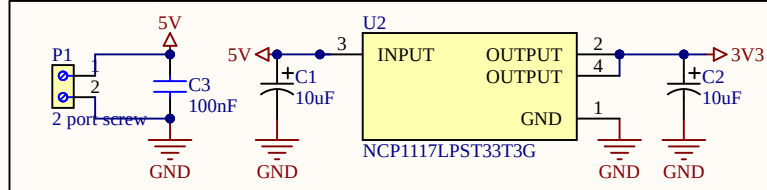
## Primary Flash and Boot Button



## Crystal Oscillator

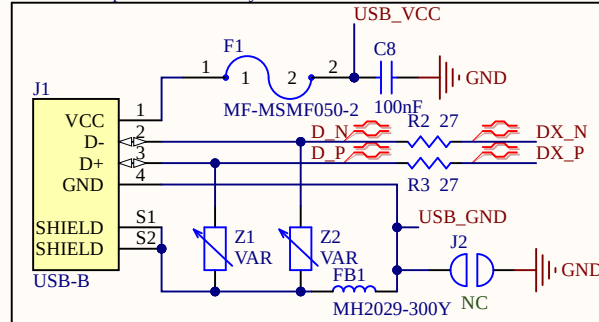


## External voltage regulator [5v =&gt; 3v3]

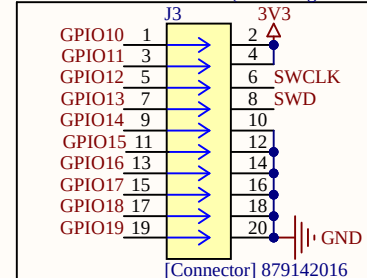


Assume a clean 5V will be provided to the board

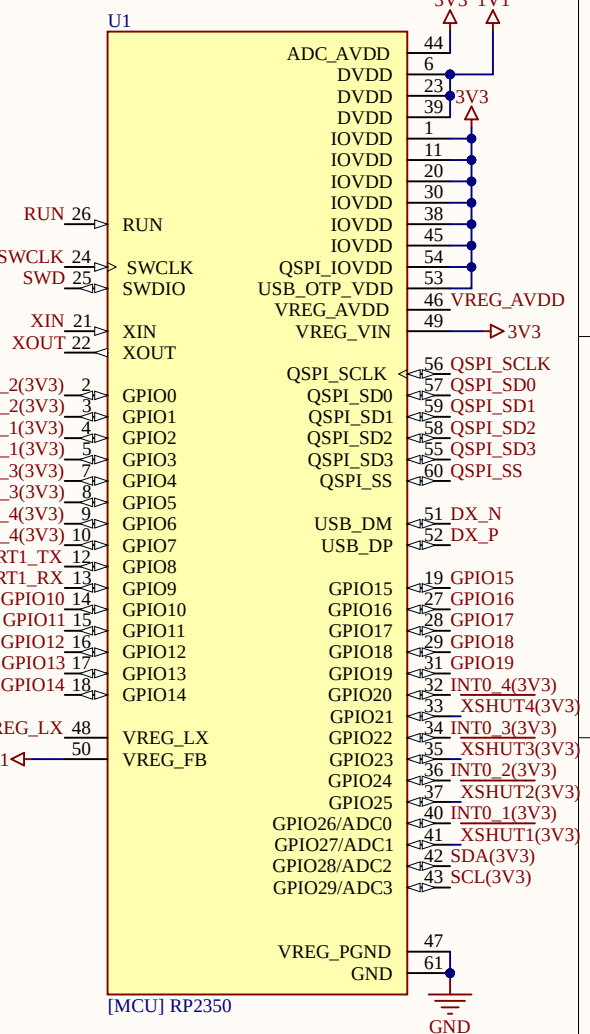
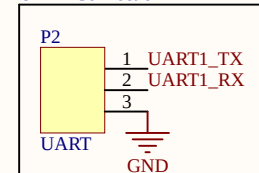
## USB-B with protection circuitry



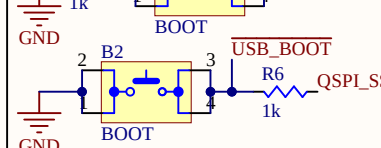
## 2x10 Extra GPIO Breakout (and debug connections)



## UART Connection



## Run and Boot Buttons



Project/Vehicle: Distance Sensor Interface

Author(s):  
-Aiden Shepler

.\*  
.\*  
.\*

Revisor(s):  
.\*  
.\*  
.\*

MIL  
MCU setup according to RPi's  
hardware design guidelines

Git Repo: \*  
Git Hash: dcac3b25

Date: 2/1/2026

Revision: 0

Size: A

File: MCU.SchDoc

Sheet 1 of 2

A

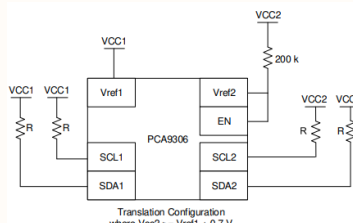
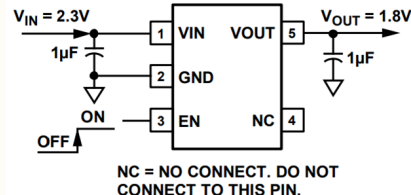
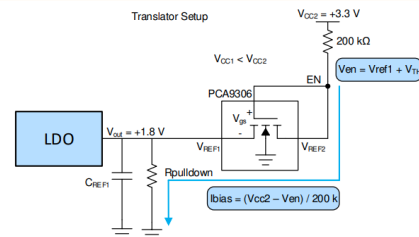


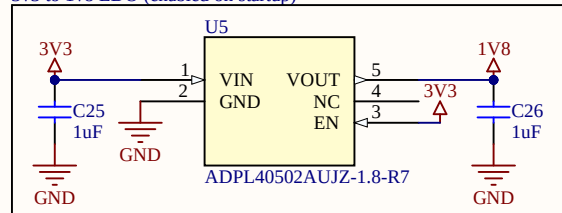
Figure 8-6. Translation Configuration



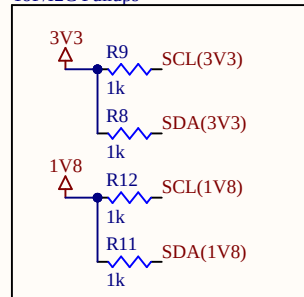
001



## 3v3 to 1v8 LDO (enabled on startup)

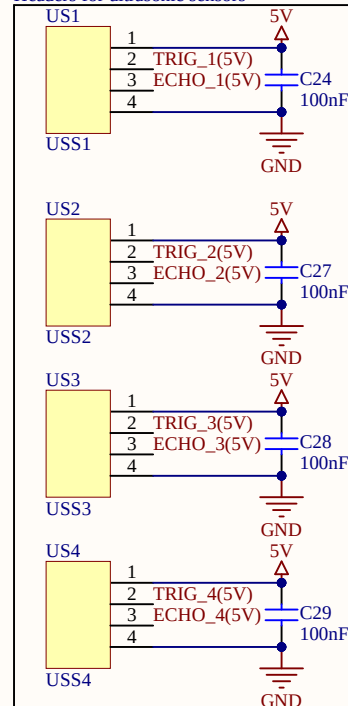


## ToF/I2C Pullups

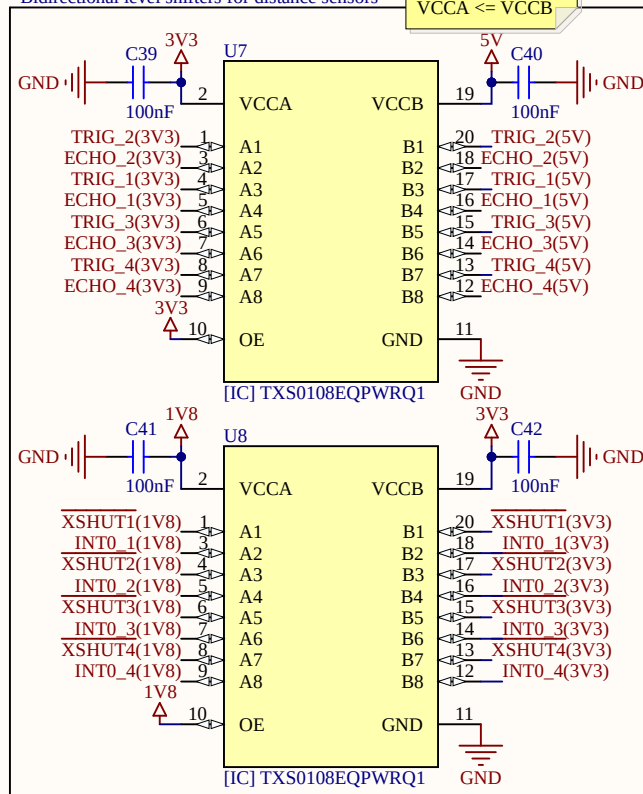


Note: GPIO1/INT0 and XSHUT pins for the ToF are pulled-up on the ToF breakout board

## Headers for ultrasonic sensors

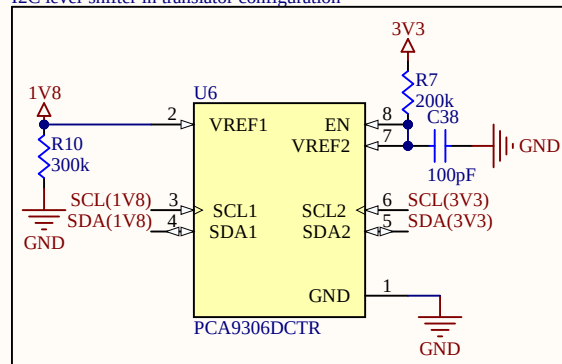


## Bidirectional level shifters for distance sensors



B

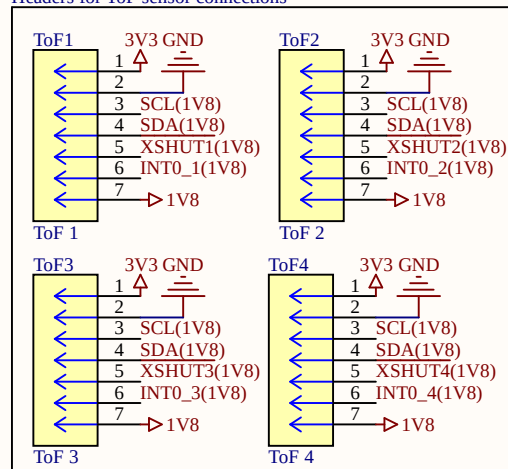
## I2C level-shifter in translator configuration



C

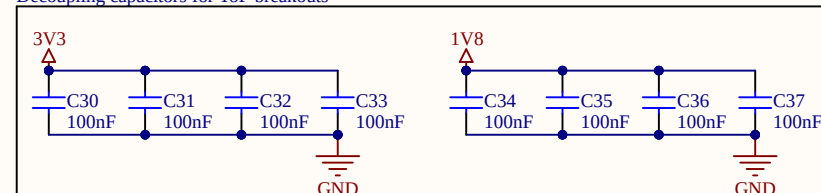
See the wiki page for the pulldown resistor calculation.

## Headers for ToF sensor connections



D

## Decoupling capacitors for ToF breakouts



Project/Vehicle: Distance Sensor Interface

Author(s):  
-Aiden Shepler  
-\*  
-\*  
-\*

Revisor(s):  
-\*  
-\*  
-\*  
-\*

MIL

Level-shifting circuitry for ToF and headers for both ToF and ultrasonic sensors

Git Repo: \*

Git Hash: dcac3b25

Date: 2/1/2026

Revision: 0

Size: A

File: Distance-Sensors.SchDoc

Sheet 2 of 2



A

B

C

D

### Time of Flight Sensors

Tof4

IVE

# Board Stack Report